

Logic:

The steps /solutions required for solving the problems known as logic.

Types of logic:

Following are three different types of logic

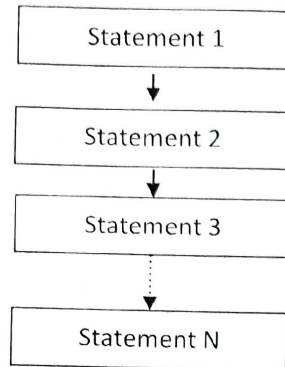
- 1) Sequence Logic
- 2) Selection(Decision making) Logic
- 3) Iteration Logic

Let's see all these logics briefly...

1) Sequence Logic:

In sequence logic all instructions or statements are executed in sequence manner as they appear. That is from top to bottom, one after another in sequence manner.

Working of sequence logic is shown in following fig:



e.g Following example uses the sequence logic

step 1: Start

step 2: read the value of a

step 3: read the value of b

step 4: calculate $c=a+b$;

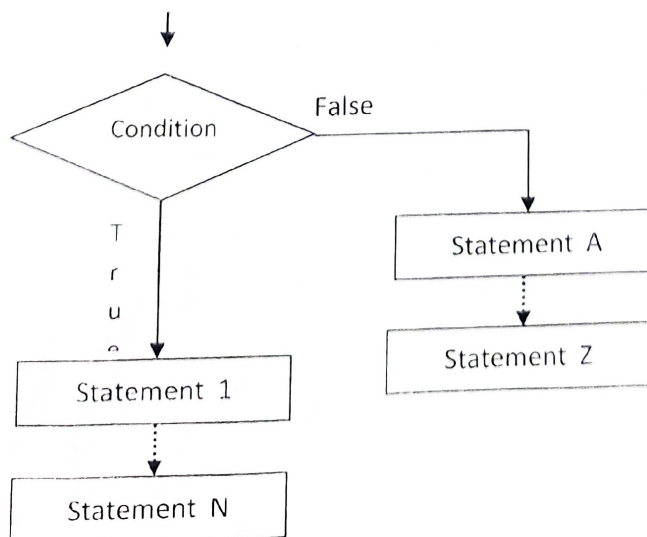
step 5: print c

step 6: Stop

1) Selection (Decision making) Logic:

Selection logic is also known as "Decision Logic" which is used to make certain decision. To solve problem, in some situation we have to check condition for that selection logic is used. This logic selects or chooses correct option or choice from number of choices. It uses If-Then, If-Then-Else or switch statements.

Working of selection logic is shown in following fig:



e.g. Following example uses the selection logic.

1) Write the algorithm to check entered number is positive or negative.

Step 1: Start

Step 2: Read N

Step 3: If ($N > 0$) Then

Display 'N' is Positive

Else

Display 'N' is Negative

Step 4: Stop

2) Iteration Logic (Looping):

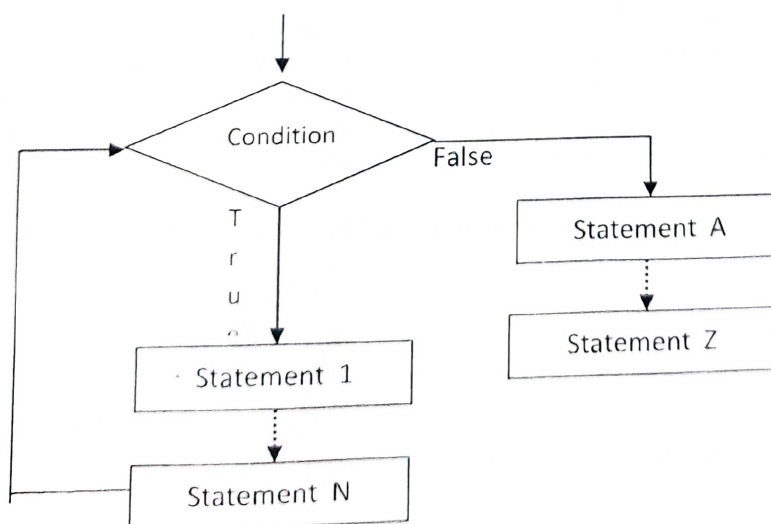
Iteration logic is also known as "Looping". In some situation we have to repeat some statements again and again at that time Iteration logic i.e. looping is used.

Looping:

"It is the process to execute some statements repeatedly until the condition becomes false."

The looping is continue as long as the condition is True and stops when condition becomes false. This logic uses Do-while, while, for, Repeat-until statements.

Working of Iteration logic is shown in following fig:



e.g. Following example uses the Iteration logic.

1) Write the Pseudo code to Print series like 1 2 3 4 5.

Pseudo Code:

Set 1 to N

While ($N \leq 5$)

Display N

Add 1 to N

End While

Stop

3) Data Type:

Data: "Data is nothing but collection of raw information or unprocessed information that we provide for the computer for processing"

e.g. numbers, string, alphanumeric etc.

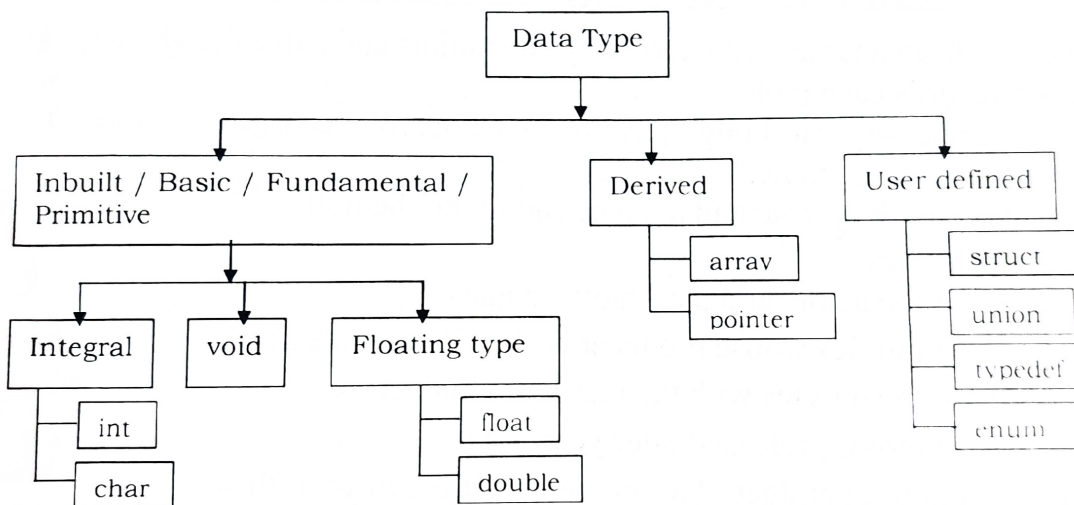
Data Type:

Concept: When we give data to the computer for processing at that time compiler does not know which type of input data is.

Generally, Data types are used to tell the compiler which type of input data is.

Definition: "Type of Data is called as Data Type"

Following figure shows data types in 'C' language:



The 'C' language supports for five fundamental or Basic or Inbuilt or primitive data types namely int, char, float, double and void.

Let's see all these basic data types in details:

1) int data type: (Integer data type)

If we want to store integer value into variable then we have to declare the data type of that variable to int

Integers are whole numbers with the range value -32768 to 32767. Also to store one integer value 2 bytes (16 bits) of memory is required.

e.g. int x;
 Here;
 int data type is used to tell the compiler that 'x' is a variable and it only stores integer value.

2) char data type: (Character data type)

If we want to store single character into variable then we have to declare the data type of that variable to char. The character variable having the range value -128 to 127. Also to store one character 1 byte (8 bits) of memory is required.

e.g. char x;
 Here;
 char data type is used to tell the compiler that 'x' is a variable and it only stores single character.

3) float data type:

If we want to store floating value (the value having decimal point) into variable then we have to declare the data type of that variable to float. Floating numbers are real numbers having decimal point with the range value $3.4e-38$ to $3.4e+38$. Also to store one float value 4 bytes (32 bits) of memory is required.

e.g. float x;
 Here;
 float data type is used to tell the compiler that 'x' is a variable and it only stores floating value.

4) double data type:

If we want to store floating value into variable then we can also declare the data type of that variable to double. Double numbers are also real numbers having decimal point with the range value $1.7e-308$ to $1.7e+308$. Also to store one double value 8 bytes (64 bits) of memory is required.

e.g. double x;
 Here;
 double data type is used to tell the compiler that 'x' is a variable and it only stores one double type value.

5) void data type:

The void is also basic data type in 'C' language. The void means no value. Generally void data type is used to specify return type of function. If we specify return type of function is void then that function does not return any value.

4) Identifier:

"Identifier is the name given by the user or programmer to the variable or pointer or array or any function"

While giving name we have to follow some rules and they are as follows (Rules to declare Identifier):

- 1) Identifier should not be keyword.
- 2) Identifier must begin with alphabets or underscore (_)
- 3) Identifier should not start with digit.
- 4) Identifier does not contain white space characters such as horizontal tab, vertical tab, space or new line etc.
- 5) Identifier contains the combination of alphabets (Small or Capital) and digits.

5) Variable:

"Variable is the name given to the memory location where the data is stored such quantity is called as Variable"